

UVTT V2 UPGRADER WORKSPACE

The Game Master's Binder Reference Booklet

A Practical Blueprint and Structural Guide to Upgrading Legacy Maps to Dynamic VTT Specifications

Welcome to your physical reference booklet for the **UVTT v2 Upgrader Web SPA**. This booklet translates the system's high-performance, CAD-style workspace into practical blueprints. It is designed to sit in your GM binder so you can quickly configure walls, portals, lighting, spatial soundscapes, and automation links during offline map design.

1. CORE RUNTIMES: WEB VS. DESKTOP PRO

While both environments share the exact same UI and drafting tools, they handle your computer's hardware resources very differently. Choose the appropriate runtime based on your campaign's map scale and resource complexity:

Feature Parameter	Web Application (Browser SPA)	Desktop Pro (Native Client)
File Ingestion Method	Manual drag-and-drop or file upload into browser window due to strict HTML5 security protocols.	Natively mounts your local hard drive OS folder; select your master asset folder once to link everything.
Streaming & RAM Scale	Constrained by strict browser-enforced memory caps. Best suited for quick edits or single-layer maps.	Natively streams gigabytes of high-resolution textures, tokens, and audio directly from your local drive.
Optimal Use-Case	Single-floor dungeons, quick touch-ups, or editing on-the-go on any machine.	Massive multi-story megadungeons, city maps, and heavy spatial audio soundscapes. Prevents RAM crashes.

2. SCENIC TOUR OF THE WORKSPACE

The interface is mapped into five dedicated zones to keep the drafting layout completely clean:

- **The Canvas Viewport (Center):** High-performance, WebGPU-accelerated rendering canvas where your map lives. Floating in the bottom right corner is the *Toggle Grid Coordinate Cursor* button, tracking active mathematical X/Y coordinates.
- **The Top Bar (Active Floor & History):** Anchored at the top center. Houses the floor switcher (globe icon & dropdown), map layer renamer, Undo/Redo controls, and buttons to add blank layers, import legacy maps (.dd2vtt/.df2vtt), or delete floors.
- **The Global Panel (Top Right):** The master command center. Houses collapsible menus for File & Export, Map Levels, Format & Platform Exports, Action History, Environment Config, and Global Audio.
- **The Export Manager (Bottom Right):** Contains the Export Profile dropdown and the green *Run Compile & EXPORT* button to build final game packages.
- **The Left Drafting Toolbar:** Houses specialized sub-panels containing your drafting toolkit: *Architecture* (structural), *Entities* (interactive), *Assets* (Desktop Pro mounted library), and *Environment* (visual test overlays).

II. THE STRUCTURAL TOOLKIT (ARCHITECTURAL DRAFTING)

Architectural drafting tools define the boundaries of exploration. Line geometries, collision zones, and overhead canopies are constructed using coordinates and mathematical snapping rules.

1. THE WALL TOOL (PHYSICAL GEOMETRY & VISION)

Walls block or guide lines of sight and token movement. Selecting the tool with no wall selected configures placing defaults, while clicking an existing wall activates Editing Mode, revealing clipboard utilities (Copy, Paste, Duplicate, Delete) and Player View visibility overrides.

- **Apply Smoothing Pass:** Mathematically curves linear walls. *Note: This utility only enables if the selected wall contains at least three vertices (two segments). Single straight lines cannot be curved.*
- **Collision Presets:** Instantly define how tokens and light interact. Use *Solid* for standard stone walls, or *Ethereal* for low fences, illusory walls, or lines that block sight but allow token movement.
- **Directional Blocking (One-Way Walls):** Define vision blocking that works from one side only. This is configured using the *Right-Hand Rule*: vision is blocked looking from the left of the line's drawing direction to the right, or vice-versa, based on drawn vertices.
- **Z-Height Bounds:** Define the exact vertical floor (bottom) and ceiling (top) elevations of the wall in grid units to support complex, multi-level verticality.
- **Snap All Vertices:** Instantly aligns wall coordinates to the active grid. Can be run globally or on selected walls in Editing Mode to prevent dynamic light leaks on your virtual tabletop.

2. THE PORTAL TOOL (DOORS, WINDOWS, & PASSAGES)

Portals define structural elements that can change state during gameplay (opening, closing, locking). You can bulk-edit selected portals to swap types or properties instantly.

- **Portal Architecture:** Dropdown to define the physical structure: *Door* (standard player toggle), *Window* (blocks movement but permits light and vision), *Secret Door* (invisible to players until discovered), or *Portcullis*.
- **Initial State:** Sets how the entryway spawns when loaded in the virtual tabletop: Open, Closed, or Locked.
- **Z-Height Bounds (Elevation Ranges):** Perfect for placing high dungeon slit windows, arrow slits, or elevated cage grates that only block sight and movement at specific elevations.

3. THE ROOF TOOL (OVERHEAD CANOPIES & MULTI-STORY LAYERS)

Roofs define overhead bounding polygons (such as forest canopies, ceiling covers, or second-story floors). In compliant Virtual Tabletops, these zones dynamically fade or completely hide the moment a player's token steps underneath them.

- **Roof Tint/Color & Opacity (%):** Define how the roof renders for the GM (or on the map before tokens step inside). Adjust opacity to make tree canopies translucent or roofs fully opaque.
- **Z-Height Bounds:** Defines the top and bottom limits of the overhead canopy (essential for multi-tier towers where players can stand on top of a roof or run beneath it).
- **Hidden from Players:** Keeps the overhead zone invisible to the players while keeping it fully visible and manageable for the GM.

4. GRID ALIGNMENT: THE 'RUBBER SHEETER'

Aligning baked-in map art grids with VTT coordinate systems is known as Rubber Sheeting. Selecting the Grid Alignment tool opens a calibration panel providing both manual and automated methods for perfect calibration:

The Automated 3-Step Workflow (Highly Recommended):

- **1. Pin Origin:** Click the top-left intersection of any grid square on the raw map artwork to lock the mathematical origin point.
- **2. Drag Auto-Align Box:** Click and drag a green bounding box over a single 1x1 map square on the artwork.
- **3. Calculate:** The Upgrader automatically calculates the optimal pixel-per-grid-unit DPI and offset coordinates.

Manual Fine-Tuning Calibration:

- **Grid X/Y (DPI) & X/Y Offset (px):** Input boxes to manually scale (DPI) or shift the grid starting position along horizontal or vertical axes.
- **Micro-Nudge Buttons:** Click → **Step +X**, ← **Step -X**, ↓ **Step +Y**, or ↑ **Step -Y** to nudge the grid by single-pixel increments.

III. THE INTERACTIVE TOOLKIT (DYNAMIC ENTITIES & TRAPS)

Entities are placing anchors that bring the environment to life with light, sound, spawn points, and mechanical automation. When you select any active entity on the canvas, the system prepends universal CAD controls: Clipboard (Copy, Paste, Duplicate, Delete), Visibility Overrides (force show/hide in Player View), and Quick Actions (Clone/Delete).

1. DYNAMIC LIGHTS & SPATIAL AUDIO

Configure visual ambiance and acoustic environments using precise spatial falloffs:

The Light Tool:

- **Lighting Projection Type:** Choose between Point (360-degree omnidirectional glow) and Cone/Beam (directional headlights).
- **Z-Axis Elevation & Radius:** Define floating height in grid units, and configure Bright Radius vs. Dim Radius boundaries.
- **Hex Color, Intensity & Decay:** Select exact hex colors. Determine light fade styles using the Decay Model (Inverse Square vs. Linear).
- **Animation Profile:** Inject environmental movement with animated presets: flicker (torches), pulse (runes), or strobe (alarms).

The Audio Tool:

- **Acoustic Falloff:** Set Inner Core (radius of constant 100% volume) and Max Range (outer limit where sound fades linearly to 0%).
- **Muffled by Walls (Occlusion):** A critical checkbox. Tells the VTT engine to automatically muffle/attenuate the spatial audio if a drawn wall breaks the direct line of sight between the player token and the sound source.

2. SPAWN POINTS, EMITTERS, & GLOBAL SCHEMATICS

The Spawn Tool:

- **Token Arrival Heading:** A slider (0 to 360 degrees) to set the exact direction tokens will face upon spawning.
- **Default Landing Zone:** Mandatory Checkbox. Every map layer must have exactly one designated Default Landing Zone so the VTT knows where to place newly entering player characters.

The Emitter Tool (Particle Weather):

- **Map-Wide vs. Local:** Weather can fill the entire canvas (blizzards) or be contained within defined rooms (basement fog).
- **Weather Styles:** Particle presets including Rain, Snow, Embers, and Fog.
- **Z-Index Layering & Particle Sliders:** Set particle rendering height (above or below roofs) and use sliders for Scale (%), Direction, Speed, Intensity (Density), and Variance (Fluctuation %).

3. THE EVENT TOOL & WIRE TARGETS (THE TRAPMASTER'S BLUEPRINT)

The Event Tool builds trigger-response interactions without coding. Triggers are physically linked to targets (like locked doors, portals, or lights) using the Wire Targets system:

- **Activation Methods:** Set how the trigger fires: *On Step* (token enters zone), *On Click* (player interacts), or *GM Only* (invisible trigger operated solely from the GM overlay).
- **Target Actions:** Determine what happens when fired: Teleport, Toggle Visibility, or Play Sound.
- **Establishing the Wire Link:** Select both the Event Trigger and the Target Object (e.g., a Door) on the canvas. The Wire Targets panel will slide open in the floating panel. Click **Bind Selected Entities** to establish the wire link. To dismantle the connection, click **Clear Targets**.

4. ADVANCED WORKFLOW: MULTI-LEVEL TELEPORTATION & STAIRS

When building complex, multi-floor dungeons, wiring up staircases or teleporters manually can be a tedious process of bouncing back and forth between map layers. The UVTT v2 Upgrader automates this through the **Teleport Routing** sub-panel.

When you select the **Event Tool** and set the Event Type to either "**Teleport**" or "**Stairs/Ladder**", the panel dynamically updates to reveal the Destination Routing controls.

Destination Routing:

- **Target Map/Floor:** Select the destination map level from your project catalog (e.g., sending a player from the "First Floor" to the "Basement").
- **Target Landing Zone:** Once a floor is selected, this dropdown populates with all available Spawn Points on that specific destination map. Select the exact spot the player will appear.

■ Automated Two-Way Linking (Default Mode Only):

To save Game Masters time, the Upgrader includes powerful automation for two-way travel. *Note: These automation checkboxes only appear when you are configuring defaults for a new event (before you click the canvas to place it).*

- **Create return Spawn point (adjacent to Event):** When checked, placing your teleport event will automatically drop a linked Spawn Point on the grid right next to it. This ensures that when players travel *back* to this floor later, they have a designated landing zone waiting for them.
- **Auto-Create Reciprocal Return Link:** The ultimate time-saver for staircases. When checked, placing your event not only links to the destination floor, but it tells the compiler to automatically generate the exact reverse event and spawn point on the destination floor pointing back to you. You get a fully functional, two-way staircase with a single click.

WIRE TARGETS FUNCTIONAL FLOW SCHEMATIC:

```

[ PLAYER TOKEN ] --(steps on)--> [ EVENT TRIGGER (On Step) ]
                                | [ BIND WIRE LINK ]
                                v
[ TRIGGER ACTION ] -----> [ PORTAL TARGET: OPEN DOOR ]

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IV. GLOBAL LAYERS, CACHE MANAGEMENT, & EXPORT WORKFLOWS

1. MAP CATALOG & FOUNDATIONAL GRID VARIABLES

Configure underlying grid mathematics and layer structures:

- **Grid Topology:** Select grid layouts: standard Square, hexagonal grids (Hex Horizontal / Hex Vertical), or Isometric layouts.
- **Grid Size & Color:** Explicitly define pixel-per-grid resolutions (e.g., 70px or 100px) and customize visual grid colors.
- **Main & Subgrid Thickness Sliders:** Separately adjust visual weights of macro grids (5-foot movement squares) and micro grids (used for high-precision vertex snapping).
- **Map Levels (The Catalog):** View multi-floor layers in a structured list. Rename layers with the Pencil (Edit) icon, or permanently purge structures using the Trash Can (Delete) button.

CRITICAL RAM SAVIOR (Action History Cache Control):

The Upgrader saves a complete snapshot of your map's data with every single edit to feed its robust Undo/Redo state machine. During long, detailed drafting sessions, this state history will consume massive amounts of your web browser's allocated RAM, leading to out-of-memory crashes.

GM Action: Open the Global Panel's Action History and click the **Clear History** button. This flushes the undo cache, instantly freeing browser memory without affecting your current active map layout. Flush the cache every 45-60 minutes during heavy structural editing.

2. THE EXPORT MANAGEMENT WORKFLOW

Compiling translates your working database into VTT-readable files. Use these options in the Global Panel and Export Manager to package your project:

- **Save & Load Project:** Always save your raw project file. This file preserves raw Bezier curve vectors, original textures, and event wiring so you can re-load and edit your layout in future sessions.
- **Package Catalog as Compound Dungeon:** Checkbox. When checked, the compiler bundles all floors in your catalog into a single .uvtt2z ZIP archive and automatically rewrites all teleporter links to route internally. When unchecked, it exports levels in *Federated Mode* as independent files.
- **Format & Platform Compile Rules:**
 - *Export v1 (Downgrade):* Converts modern v2 maps back into legacy .dd2vtt structures.
 - *Compile v2 (Standard):* Standard compilation to the Universal Virtual Tabletop v2 specification.
 - *Secure Archive (.zip):* Compiles your project into a secured .uvtt2z ZIP, decoupling structural data from premium assets and generating cryptographically secure manifest.hash files to protect intellectual property.
 - *Platform Exports:* Tailors output format for Foundry VTT, Roll20, or Fantasy Grounds.
- **The Export Manager (Bottom Right):** Set your target profile using the Export Profile Dropdown, then hit the prominent green **Run Compile & EXPORT** button to download your playable map file.